



Welcome to CSE 598/CEN 598/CSE 494: Advanced Digital Design and Verification

Fall 2026



About your instructor

Aman Arora
Assistant Professor

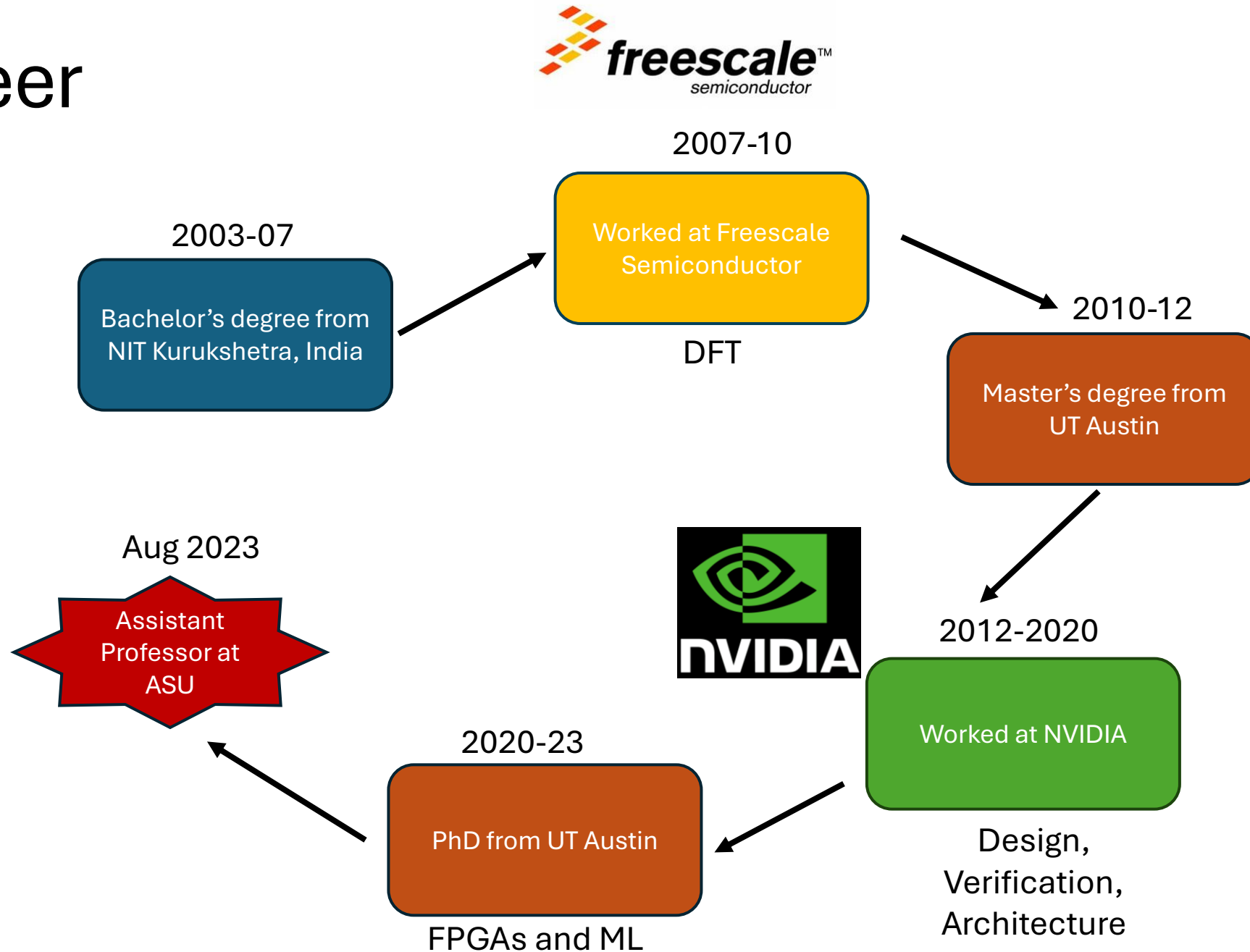
Research Interests:

- Reconfigurable Computing
- Accelerating Machine Learning (ML) applications
- ML for Hardware and CAD
- Energy-efficient computing (e.g. sustainable computing, PIM)

Website: <https://labs.engineering.asu.edu/advent/>

Associations: Advent lab, IDEAS center, ACME center

Career



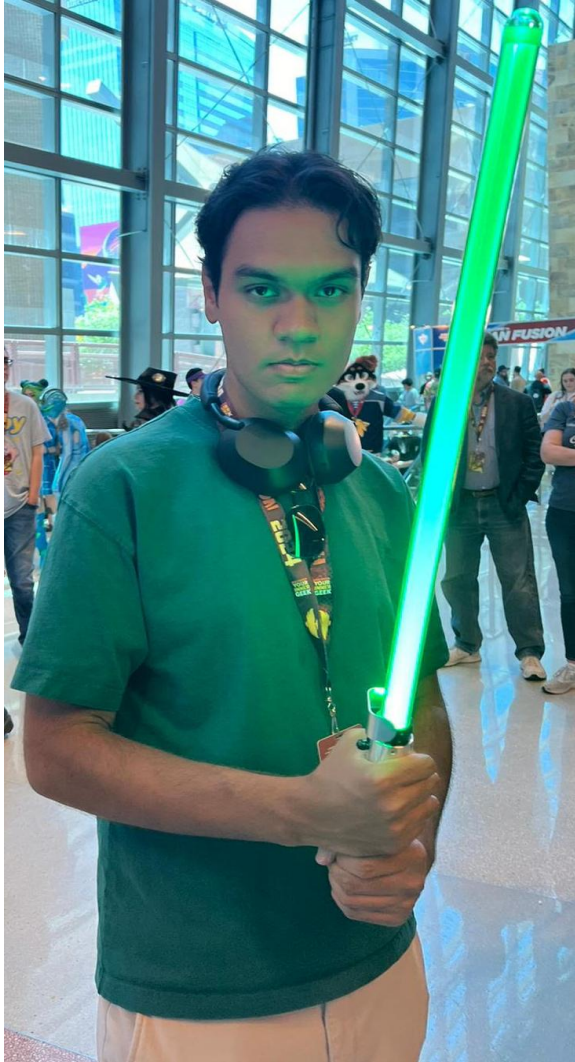
Teaching Assistant



Ruthwik Sunketa

- Degree: Ph.D. (CEN)
- Born and Raised: Hyderabad, India
- Work Experience: 6-month internship in Systems Validation at Medtronic
- Hobbies: Puzzles and Audiobooks

Grader



Hussain Attyah

- Degree: BSE (CSE)
- Born and Raised: Dubai, U.A.E
- Research Experience: Provenance Chain for USB
HIDs, Zoned Namespace SSDs & KVCache
- Hobbies: Lego, comics & coffee

Course Website

<https://adventlab.notion.site/addv-fall2026-website>



ADDV Fall 2026 Website

-  [Syllabus](#)
-  [Schedule](#)
-  [Lab Assignments](#)
-  [Background Knowledge](#)
-  [Vendor Trainings](#)
-  [Tips, Tricks, and FAQs](#)
-  [Resources](#)

Important Things to Mention

- This course is ASIC based. No FPGA experience is required.
- No hardware/boards required. Simulation/Synthesis only using EDA tools/software. You'll access the software needed through Apporto.
- I've almost combined two courses into one. So, this is a heavy course. Expect to spend at least 10 hours each week.
- In addition to the materials discussed in the classroom, I'll assign videos to watch and documents to read before/after lectures.
- On Discord, I will have notifications set to "@ mentions only". Hopefully your questions will be resolved by the TA and other students. If not, @ me.

Course Organization and Lecture Order

- Ideally I'd like to organize the course around SoC Design and Verification

SoC Design

- CPU
- Accelerator
- Interconnect
- I/O and memory controller
- Globals (clock, reset, etc)

SoC Verification

- Verification methodology
- Unit-level verification
- Sub-system-level verification
- Top-level verification

- But the order of lectures is connected to labs. So, it may feel a bit disorganized.
- Some lecture re-ordering may happen because of guest speaker availability.

3 quizzes will be released later today.

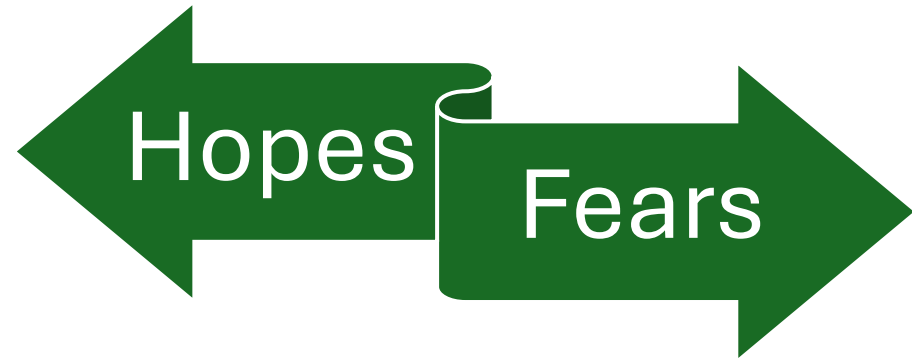
1. Kickoff Survey

2. Syllabus Quiz

3. Pre-Req Test (intentionally hard)

Let's break the ice!

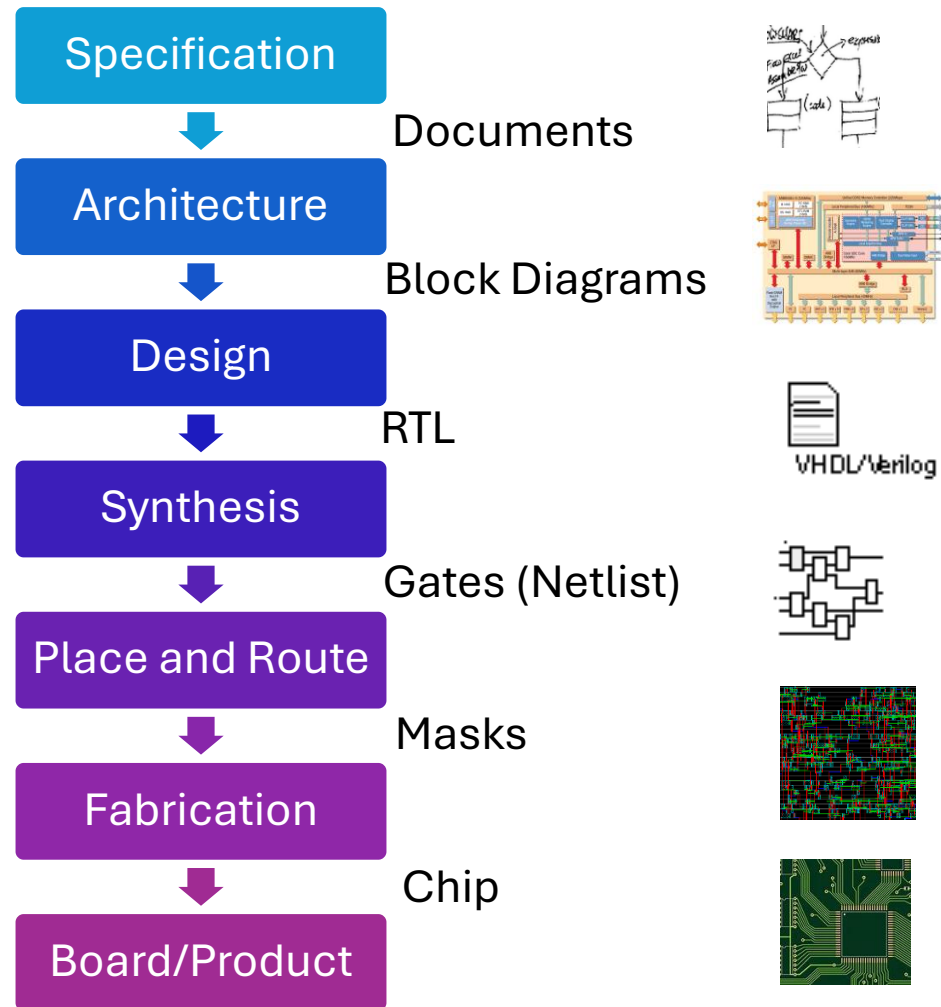
- Pair up with the person sitting behind you.
- Introduce yourself.
- Share your hopes and fears for this course.



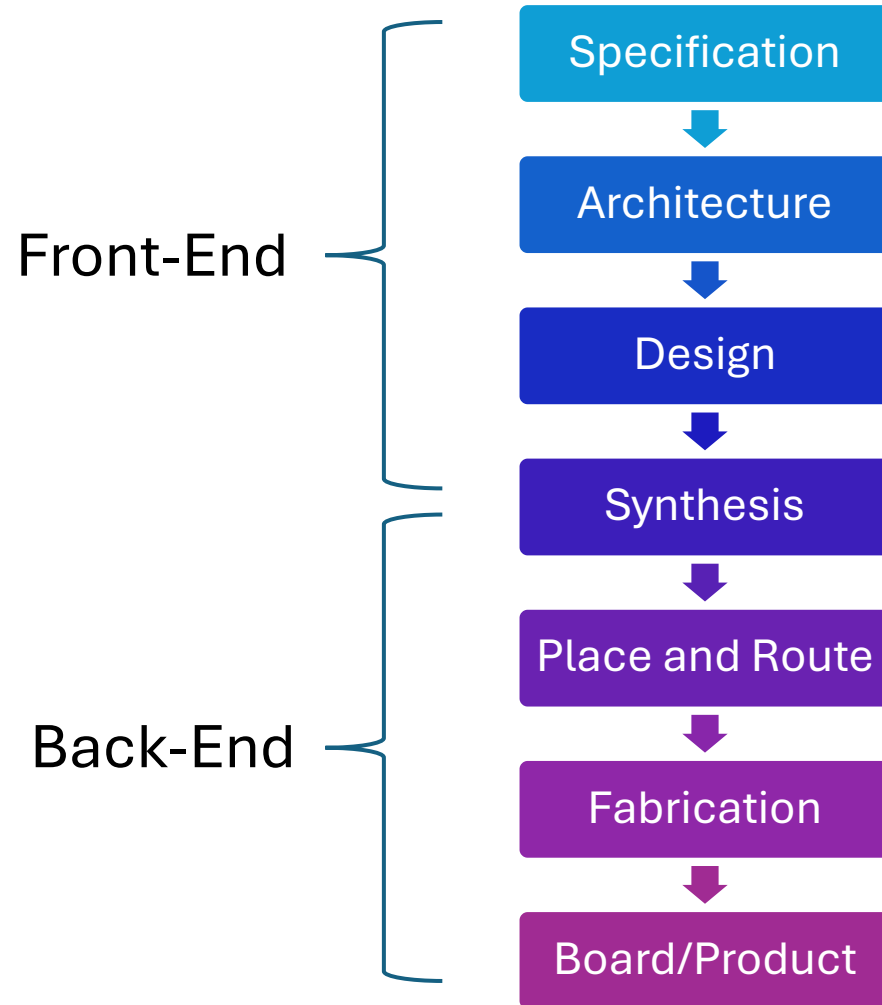
Design Flow

Design Flow

Focus of first part
of the course

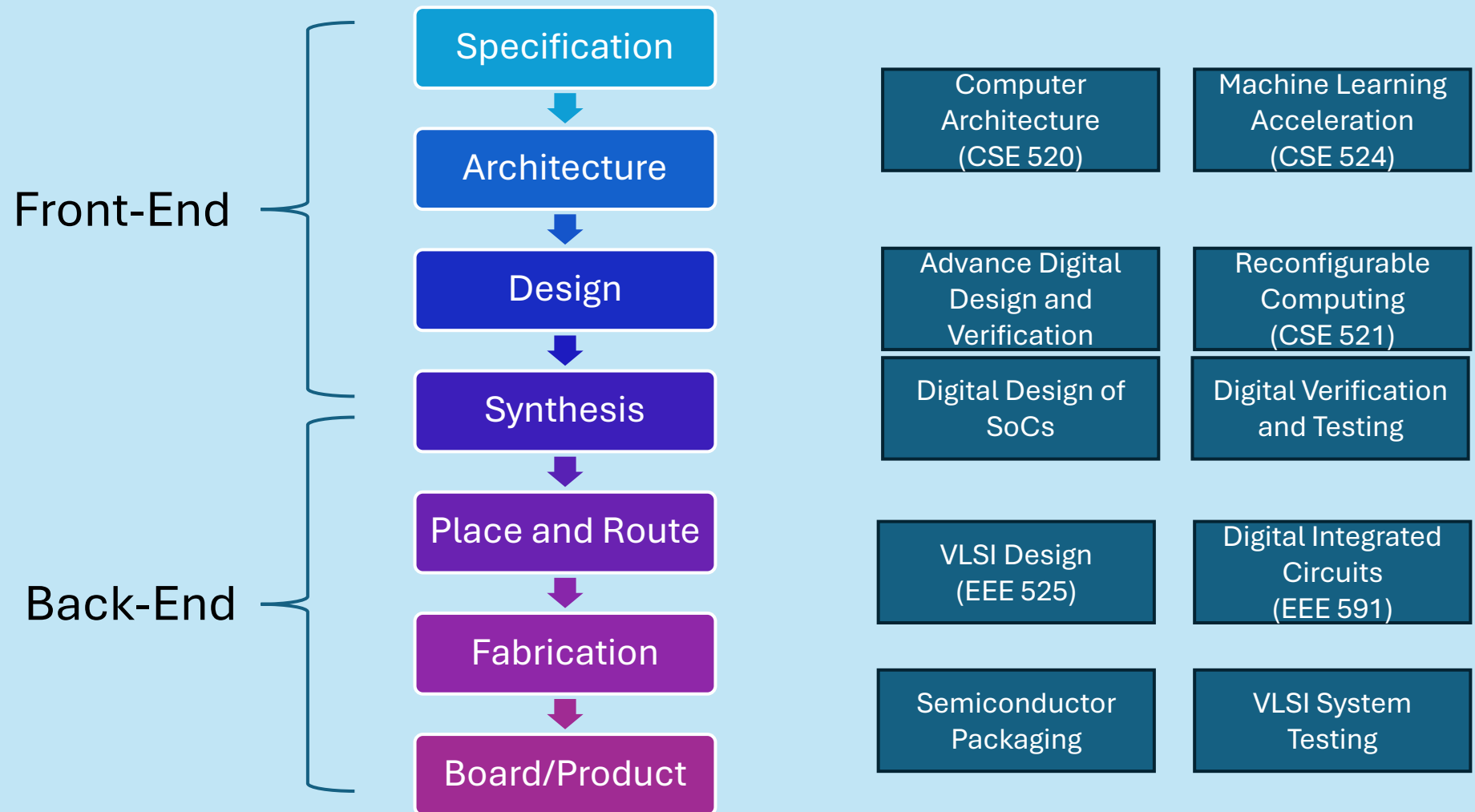


Design Flow



Design Flow

CEN/CSE Courses at ASU - Advent Lab



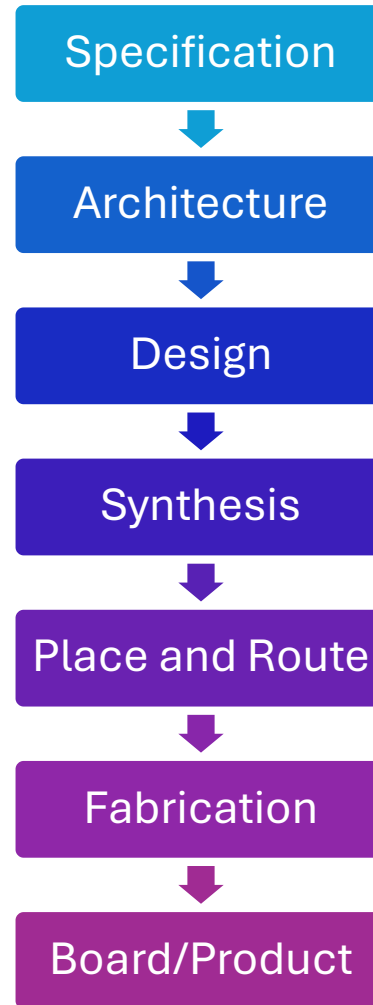
Verification is Everywhere

Focus of second
part of the course

Functional
and
Performance
Verification

Timing,
Power, Layout
Verification

Faults and
electrical
verification



Design Flow

- **Specification**

- **Goal:** Define what the system should do.
- **Details:** Functional requirements, performance targets, power constraints, and interface definitions.
- **Deliverables:** Requirement documents, use-case scenarios, timing diagrams.

- **Architecture**

- **Goal:** Define the high-level structure of the system.
- **Details:** Partitioning into modules, data paths, control logic, memory hierarchy.
- **Deliverables:** Block diagrams, interface specs, timing budgets.

- **Design (RTL)**

- **Goal:** Describe the system behavior using a hardware description language (HDL).
- **Details:** Written in Verilog or VHDL, includes simulation testbenches.
- **Deliverables:** RTL code, simulation results, synthesis constraints.

- **Synthesis**

- **Goal:** Convert RTL into a gate-level netlist.
- **Details:** Logic optimization, technology mapping, constraint checking.
- **Deliverables:** Netlist, synthesis reports (timing, area, power).

- **Place and Route (P&R)**

- **Goal:** Physically implement the design on silicon.
- **Details:** Floorplanning, placement, clock tree synthesis, routing.
- **Deliverables:** GDSII layout, timing closure reports, DRC/LVS checks.

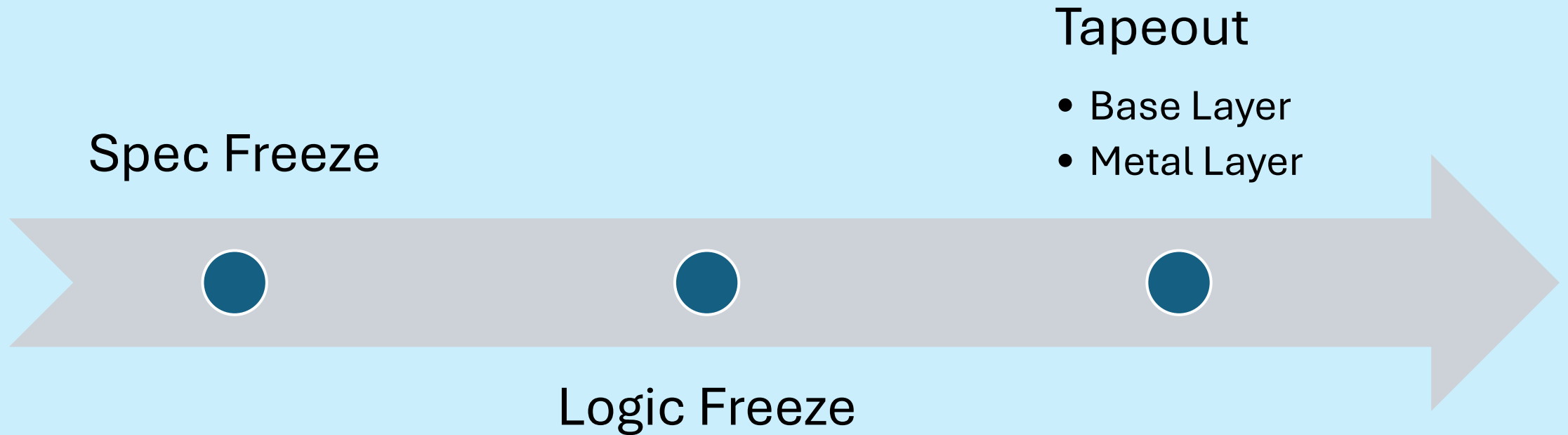
- **Fabrication**

- **Goal:** Manufacture the chip using the layout masks.
- **Details:** Sent to a foundry for wafer production, packaging, and testing.
- **Deliverables:** Silicon dies, packaged chips.

- **Board/Product**

- **Goal:** Integrate the chip into a final product.
- **Details:** PCB design, firmware integration, system-level testing.
- **Deliverables:** Final product, datasheets, user manuals.

Aside: What is an ECO (Engineering Change Order)?

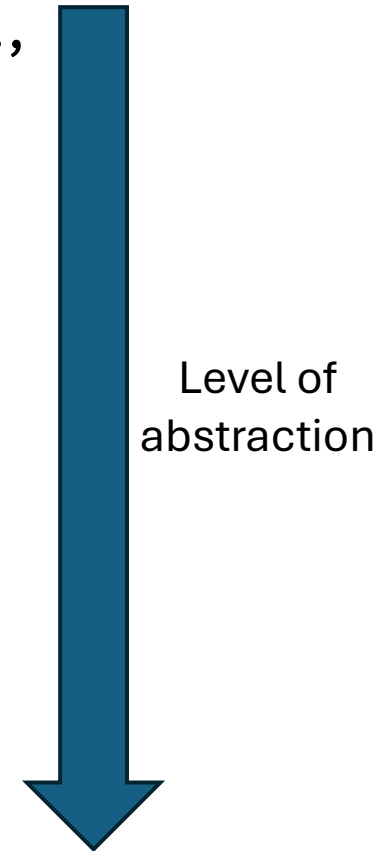


If an RTL bug is identified after Logic Freeze, it is “surgically” implemented in the netlist. Avoid re-synthesis. This is called an ECO. Approvals required.

Terminology

Architecture, Microarchitecture, Design

- **Architecture:** The abstract specification of system behavior (e.g., instruction set, communication protocol, or functional model).
 - Defines “what” operation a system performs
- **Microarchitecture:** The concrete implementation of that behavior using registers, memories, gates, control paths, and data paths.
 - Organization and structure of a digital system’s blocks
 - Defines “how” the system performs operations
 - Less abstract than architecture
- **Design:** Implementing a microarchitecture
 - Microarchitecture guides design. Serves as a “plan” for design.
 - Slightly less abstract and more detailed than microarchitecture



Verification vs. Testing

Verification

- Target design bugs
- Goal: Check that design correctly implements the specification
- Performed when the chip is being designed
- Performed once prior to manufacturing
- Responsible for quality of design
- Can observe and control all nodes in the design

Testing

- Target manufacturing defects
- Goal: Check that manufactured part correctly implements design
- Performed after the chip is manufactured
- Test application done for every chip
- Responsible for quality of shipped parts
- Can only control and observe primary inputs and outputs

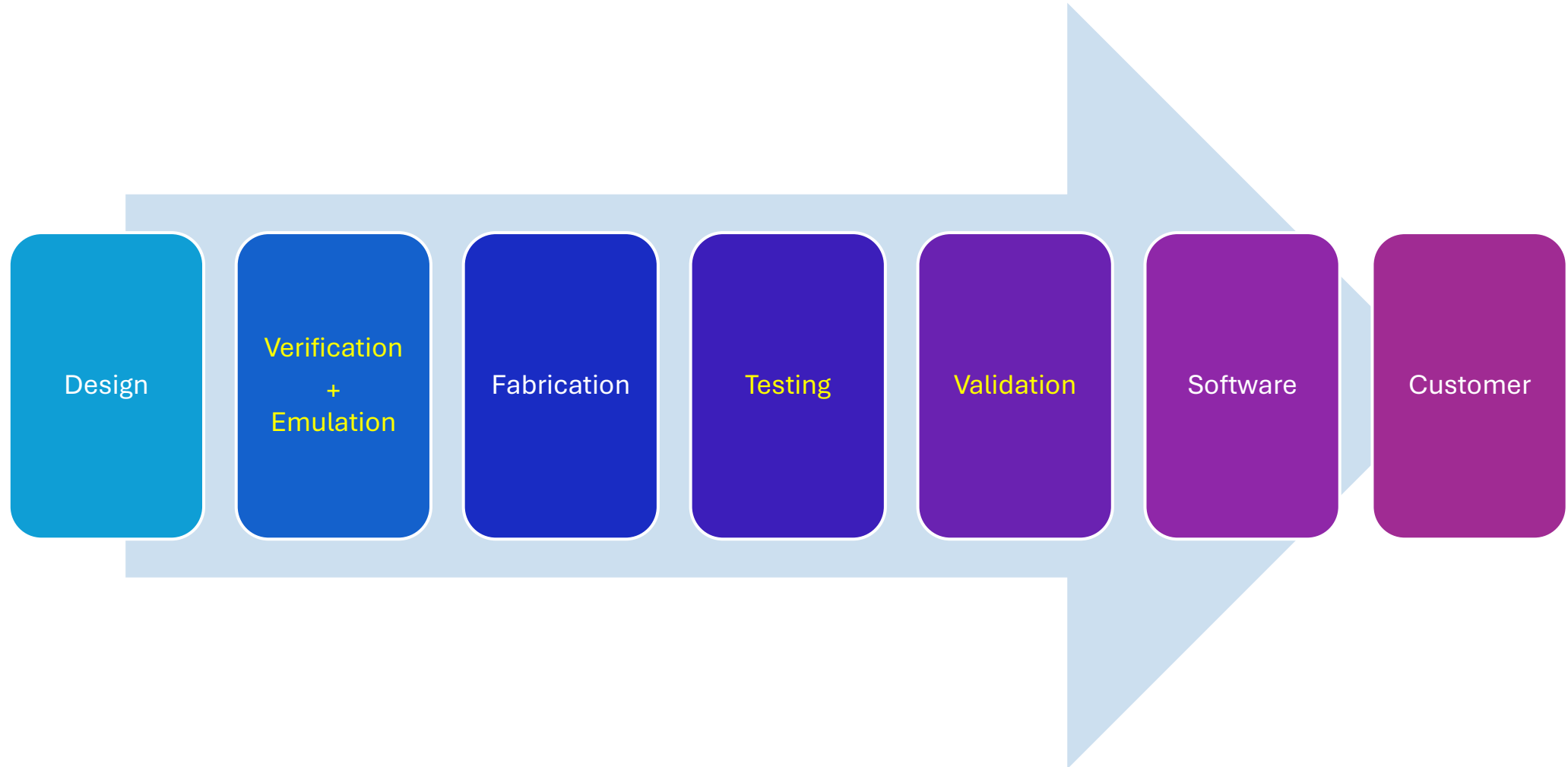
Validation

- Performed on ‘alive’ silicon
 - ie. after the device has passed ‘manufacturing test’
- Develop validation boards (EVB)
- Simple functionality checks
 - Boot the OS and run “actual” tests or benchmarks
- Perform power and electrical measurements

Emulation/Prototyping

- Using specialized hardware to realize your design before the ‘actual Silicon’
 - Either FPGA based (like *Veloce* from Siemens (Mentor Graphics))
 - Or processor based (like *Palladium* from Cadence)
- Simulation acceleration
 - Simulation is slow
- Prepares you for Validation
 - Can run software/drivers/etc before Silicon arrives
 - When Silicon arrives, re-use design/test collateral
 - Thereby, aiding Silicon bring-up
- Emulation has limitations
 - Analog blocks
 - Real clocking/reset logic
 - Real timing verification
 - Difficult to debug!!! [Simulation is slow. But you can see everything!]

What happens when?



Verification, Validation, Testing, Emulation

- Verification a.k.a. Design Verification
- Validation
- Emulation
- Testing a.k.a. Manufacturing Test

	Intel	Freescale/NXP	NVIDIA
Design Verification	Pre-Si Validation	Verification	Verification/DV
Validation	Post-Si Validation	Post-Si Validation	No specific name (“Bringup?”)
Emulation	Emulation	Pre-Si Validation	Emulation
Manufacturing Test	Testing	DFT	DFT